



Pandas: The Popular Python Library for Data Analysis and Data Science

Pandas is a popular Python library. This article describes a few features and functions available in this library, and encourages readers to use it for practical business problems.



Pandas provides fundamental and high-level building blocks for practical, real world data analysis in Python. It is one of the most powerful and flexible open source tools for data analysis and manipulation, and provides data structures for modelling and manipulating tabular data (data in rows and columns).

Pandas has two primary data structures. The first is a 'series' data structure that helps to retrieve data from the array or dictionary of Python objects. Data can be retrieved either by position or by specifying the index name. The second is the 'dataframes' data structure to store data in rows and columns. Columns have column names and rows are accessed using indexes. Columns can have different types

of data including lists, dictionaries, pandas series, another dataframe, NumPy arrays, etc.

Processing various file types

Data is often available in various formats. It is imperative that the tool used for data analysis is able to provide a wide range of methods for handling it.

With Pandas, one can read various file types like CSV, JSON, XML, Parquet, SQL (see Table 1).

Data cleansing using Pandas

In real-world scenarios, data is often incomplete and includes bad data. It is sometimes duplicated. Also, data includes sensitive and confidential

Table 1

	Write	Read
CSV	<i>to_csv</i>	<i>read_csv</i>
JSON	<i>to_json</i>	<i>Read_json</i>
Parquet	<i>to_parquet</i>	<i>read_parquet</i>
SQL	<i>to_sql</i>	<i>read_sql, read_sql_query, read_sql_table</i>
XML	<i>to_xml</i>	<i>read_xml</i>